objects passing on a conveyor, RoCycle correctly classified 27 objects with 85 per cent accuracy.

The creators believe that such robots could be used in places like apartment blocks or on university campuses to carry out first-pass sorting of people's recycling, cutting down on contamination.

RoCycle uses pincers to pick through garbage and identify what materials each bit contains. It could help reduce how much waste gets sent to landfill (Credit: MIT Technology Review)

Since the robot picks up items one by one, it is too slow for industrial recycling plants, which are expensive to run and need to process waste quickly to cover costs.

The team is working on combining its touch-based robot with a visual system to speed things up. This robot would scan objects passing by and pick up only those it was not sure about.

Blue, the human-friendly robot designed for AI

A team of researchers at University of California, Berkeley, has developed Blue, a low-cost, human-friendly robot. Blue was designed to use recent advances in artificial intelligence (AI) and deep reinforcement learning to master intricate human tasks, all while remaining affordable and safe enough so that every AI researcher—and eventually every home—could have one.


Blue, with its creators, Pieter Abbeel, David Gealy and Stephen McKinley (Credit: UC Berkeley)

Blue is the brainchild of Pieter Abbeel, professor of electrical engineering and computer sciences at UC Berkeley, postdoctoral research fellow Stephen McKinley and graduate student David Gealy. The team hopes Blue will accelerate the development of robotics for the home.

"AI has done a lot for existing robots, but we wanted to design a robot that is right for AI," Abbeel said. "Existing robots are too expensive, not safe around humans and similarly not safe around themselves—if they learn through trial and error, they will easily break themselves. We wanted to create a new robot that is right for the AI age rather than for the high-precision, sub-millimetre, factory automation age."

Blue's durable plastic parts and high-performance motors cost less than US$ 5000 to manufacture and assemble. Its arms, each about the size of the average bodybuilder's, are sensitive to outside forces—like a hand pushing it away—and has rounded edges and minimal pinch points to avoid catching stray fingers. Blue's arms can be very stiff, like a human flexing, or very flexible, like a human relaxing, or anything in between.

AI accurately predicts the useful life of batteries

Scientists at Stanford University, Massachusetts Institute of Technology (MIT) and Toyota Research Institute have discovered that combining comprehensive experimental data and artificial intelligence (AI) revealed the key for accurately predicting the useful life of lithium-ion batteries before their capacities start to wane. After they trained their machine learning model with a few hundred million data points of batteries charging and discharging, the algorithm predicted how many more cycles each battery would last, based on voltage declines and a few other factors among early cycles.

The predictions were within nine per cent of the number of